

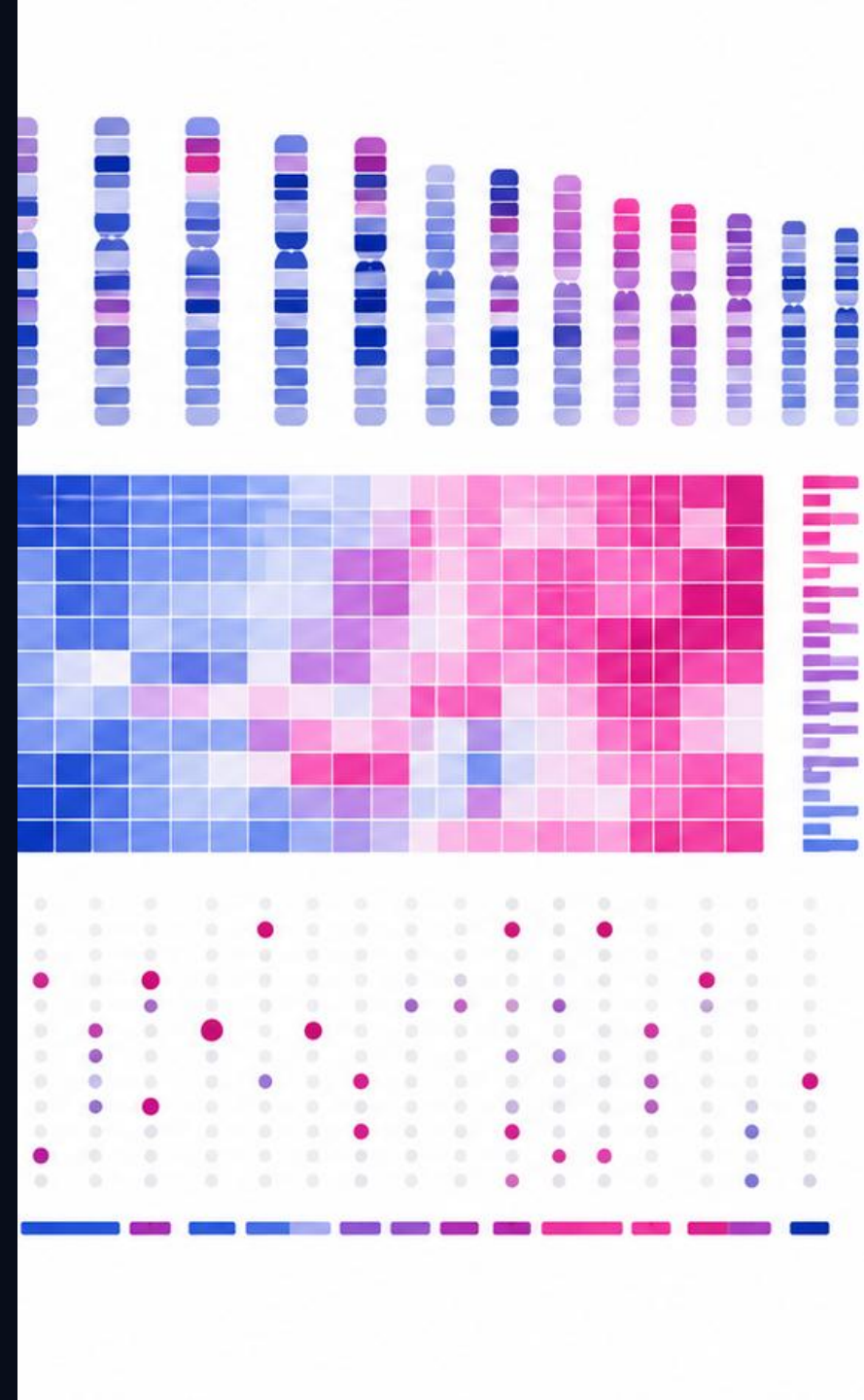
WEEK 3

Pathway Analysis & Protein Networks

Using Enrichr and STRING to move from gene lists to biological interpretation

cBioPortal gene list → Enrichr pathways → STRING network → project claim

Breast cancer genomics project



Turn a gene list into a biological story

Input

A short gene list from cBioPortal:

- altered genes
- subtype-specific genes (expression)
- candidate targets
- survival-associated genes



Interpretation

Enrichr asks:

“What pathways or functions are over-represented?”

STRING asks:

“Do these proteins form a connected functional network?”



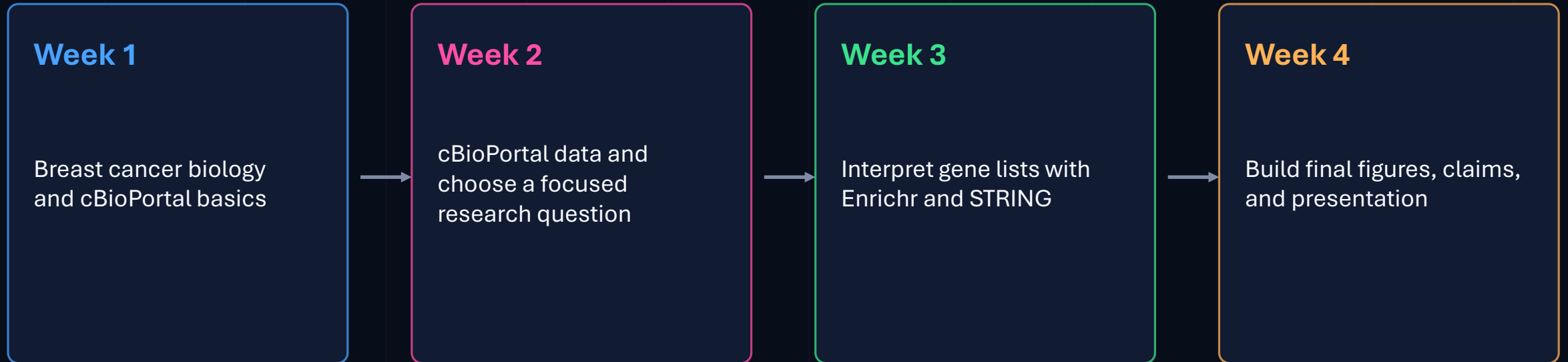
Output

A clearer project claim:

“These genes point toward DNA repair, cell-cycle control, or PI3K signaling.”

Then state limitations honestly.

Where Week 3 fits in the project



This week is not about “finding the answer.” It is about testing whether your gene list points to a biological process.

Start with a defensible gene list

Good pathway/network analysis depends more on input quality than on software settings.

Good inputs

- 10–50 genes
- collected from one clear comparison
- biologically interpretable
- gene symbols checked
- not hand-picked only because they are familiar

Risky inputs

- only 3–5 genes
- hundreds of genes with no filtering
- mixed evidence types
- copied from unrelated studies
- missing or outdated gene symbols

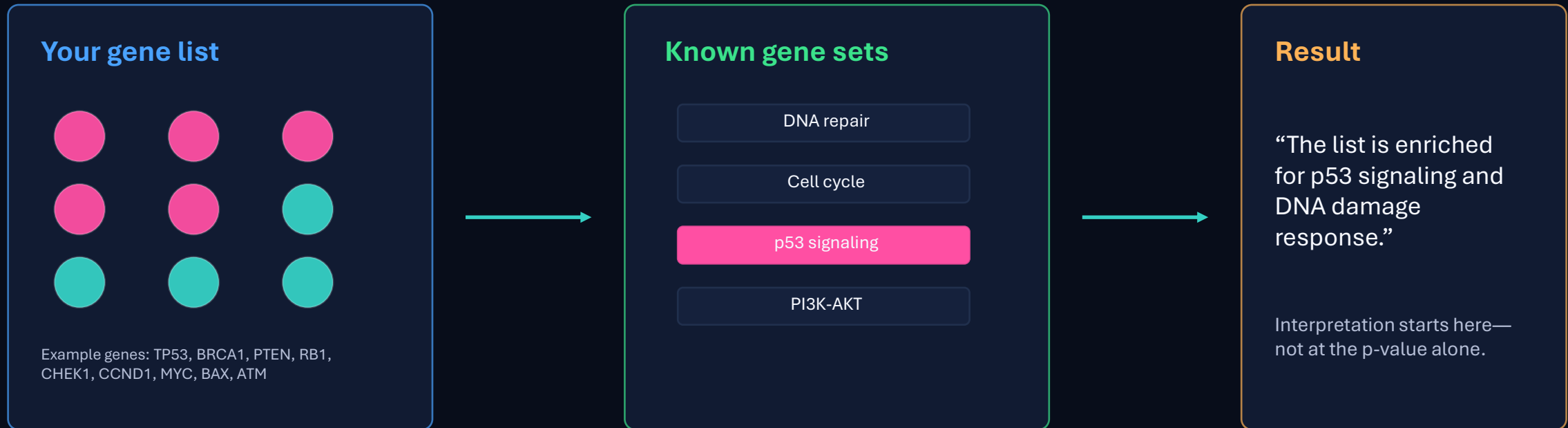
For this project

Use genes from your cBioPortal comparison:

- Genomic Alterations
- OncoPrint gene panel
- mutation-focused candidate list
- survival-associated comparison

Pathway enrichment: the central idea

Enrichment asks whether your gene list contains more members of a known pathway than expected by chance.



Enrichr: what it does

Enrichr is a web-based gene-list enrichment tool. It compares your submitted genes against curated libraries of pathways, processes, and annotations.

Use these libraries first

- GO Biological Process
- KEGG Human pathways
- Reactome pathways

These are broad enough for discovery but structured enough for interpretation.

Useful output fields

- enriched term
- adjusted p-value
- overlap genes
- combined score

The overlap genes are often the most important for your story.

Main question answered

“What biological functions are over-represented in this gene list?”

Not:

“Which pathway caused this cancer?”

Enrichr workflow

- 1 Open Enrichr**
Go to maayanlab.cloud/Enrichr
- 2 Paste genes**
One official gene symbol per line
- 3 Submit list**
Use the same list from cBioPortal
- 4 Choose library**
Start with GO Biological Process, KEGG, or Reactome
- 5 Record results**
Top terms, adjusted p-values, and overlap genes

Example input

```
TP53  
BRCA1  
PTEN  
RB1  
CHEK1  
CCND1  
MYC  
ATM
```

Use uppercase official gene symbols when possible.

What to record from Enrichr

Enriched term	Adj. p	Overlap	Genes
p53 signaling pathway	1.2e-5	5/84	TP53, CDKN1A, BAX...
Cell-cycle checkpoint	3.8e-4	4/63	RB1, CHEK1, CCND1...
DNA damage response	8.1e-4	4/97	BRCA1, ATM, TP53...
PI3K-AKT signaling	0.012	3/120	PIK3CA, PTEN, AKT1...

Record the evidence, not just the name

For each top pathway, write:

1. pathway/process name
2. adjusted p-value
3. overlap genes
4. one-sentence interpretation
5. why it matters for your project question

Interpreting Enrichr results: good claims vs. overclaims

Good interpretation

“Our Basal-like gene list was enriched for p53 signaling and DNA damage response. This supports the idea that these tumors differ from Luminal A tumors in pathways related to genome maintenance and cell-cycle control.”

Overclaim

“The p53 pathway caused the cancer, and TP53 is validated as a drug target.”

Why this is wrong:

- enrichment is associative
- databases are incomplete
- gene lists are biased by how they were collected
- target validation requires experiments

Rule: Enrichr supports a biological interpretation. It does not prove mechanism or therapeutic efficacy.

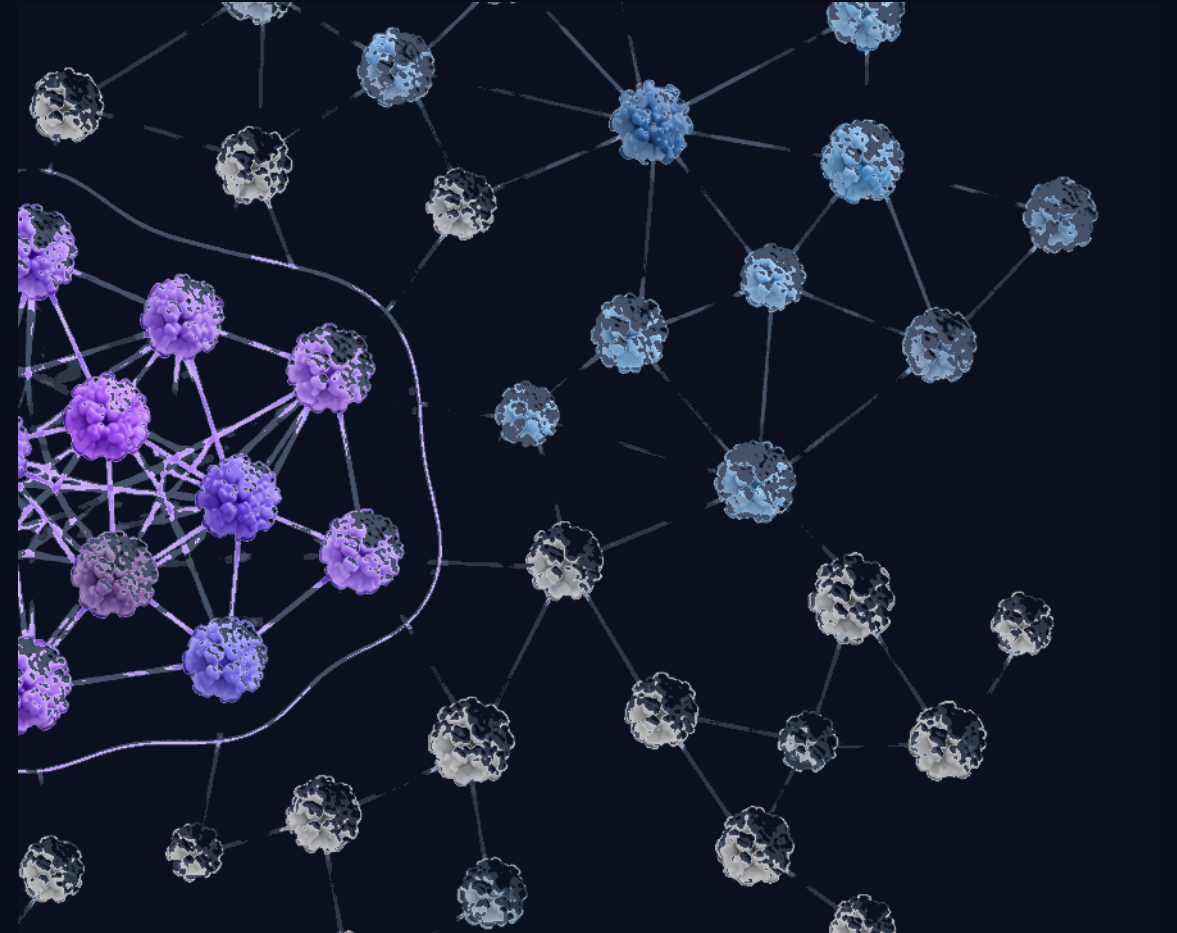
STRING: from genes to protein association networks

What STRING asks

Do the proteins encoded by your genes form a connected biological network?

STRING integrates evidence from experiments, curated databases, co-expression, text mining, and computational predictions.

Edges represent functional or physical associations, not necessarily direct binding.



STRING workflow

- 1 **Open STRING**
Go to string-db.org
- 2 **Choose Multiple proteins**
Paste the same gene list used in Enrichr
- 3 **Set organism**
Select Homo sapiens
- 4 **Run search**
Confirm mapped proteins if prompted
- 5 **Inspect network**
Look for connected clusters, hubs, and isolated proteins
- 6 **Check enrichment**
Open the Functional Enrichment section and record FDR values

Recommended starting settings

Network type:
full STRING network

Required score:
medium or high confidence

Meaning:
more stringent = fewer edges,
fewer likely false positives

If the network is too dense, increase
confidence.

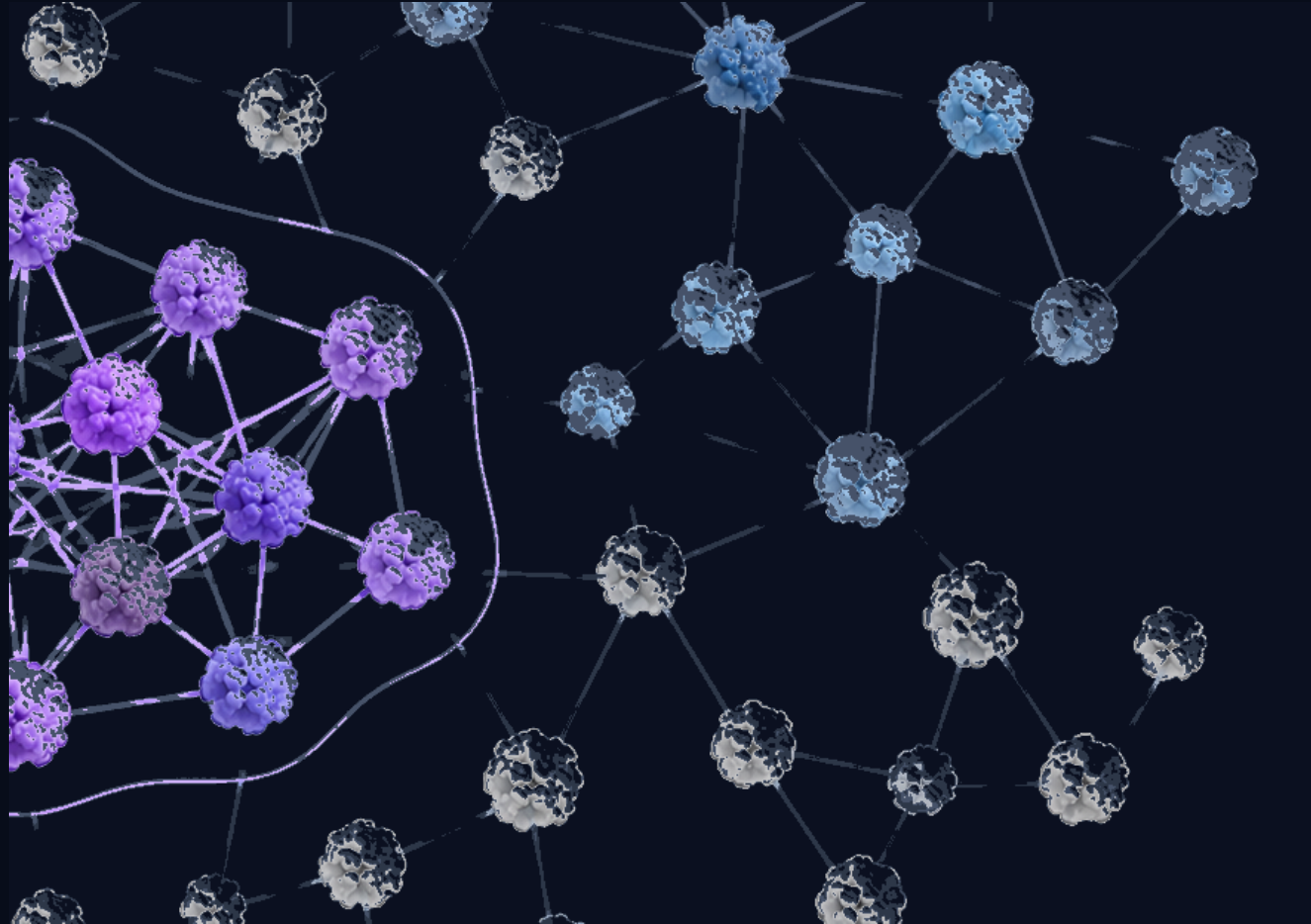
How to read a STRING network

Clusters

Connected groups may point to a shared process or pathway.

Hubs

Highly connected proteins are interesting but not automatically good targets.



Edges

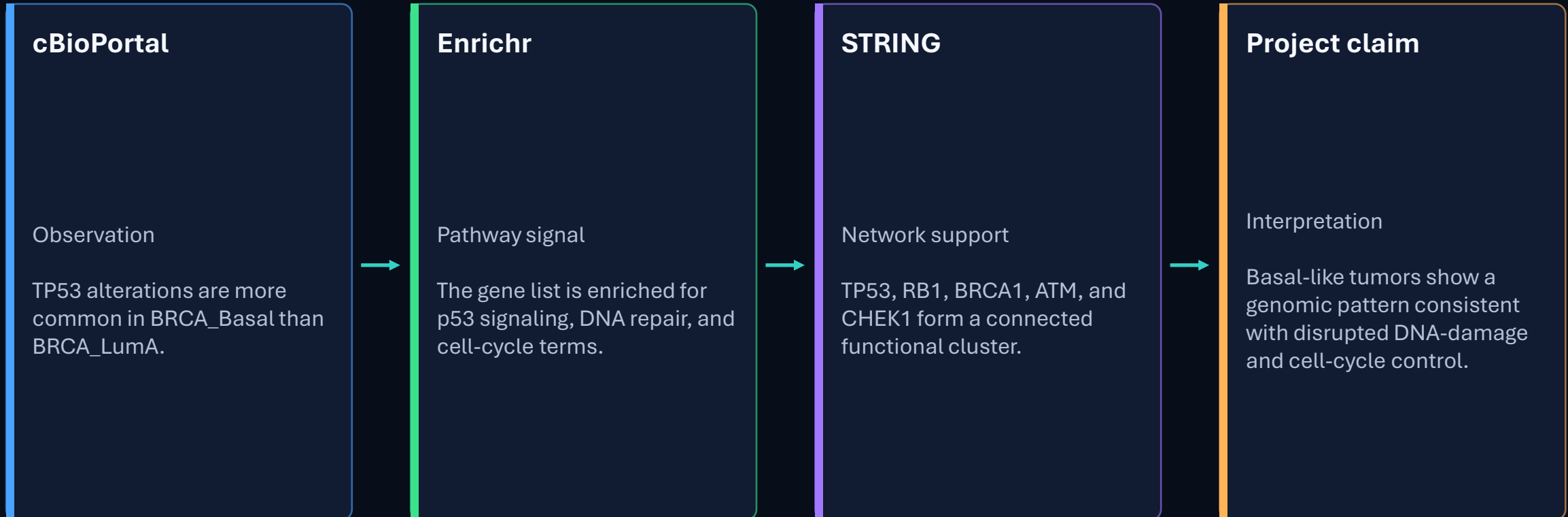
Associations; not always direct physical binding.

Isolates

Disconnected proteins may represent separate biology or weak evidence.

The network is a hypothesis generator. It helps decide what biology to discuss and what experiments would come next.

Integrating cBioPortal, Enrichr, and STRING



Notice the language: “consistent with” and “supports.” Not “proves.”

Build your interpretation

Part A: Prepare

Open your Week 2 cBioPortal results. Select a gene list of 10–50 genes from your comparison. Make sure every entry is an official gene symbol.

Part B: Run Enrichr

Paste the list into Enrichr. Use GO Biological Process, KEGG, and Reactome. Record the top 3 useful terms and overlap genes.

Part C: Run STRING

Paste the same list into STRING. Set organism to Homo sapiens. Save the network screenshot and record one cluster or hub.

Week 3 deliverables

By the end of this week, each group should have these artifacts ready for the final project.

Gene list

10–50 genes, with source: Genomic Alterations, OncoPrint, or survival comparison

Enrichr table

Top 3 useful terms, adjusted p-values, overlap genes, and interpretation

STRING network

Screenshot plus one paragraph explaining clusters, hubs, or isolates

Integrated claim

One sentence linking cBioPortal observation to pathway/network evidence

Limitations

At least two: observational data, biased gene list, cohort size, incomplete clinical data

Exit ticket: what biological process best explains your current gene list?

Source notes and cautions

Main resources

Enrichr
maayanlab.cloud/Enrichr

STRING
string-db.org

cBioPortal
cbioportal.org

Use these tools in the browser; no R or Python required.

Scientific caution

Pathway enrichment and protein networks are evidence summaries. They help generate hypotheses.

They do not prove mechanism, clinical utility, or druggability.

A strong project states what the data support and what remains untested.

Relevant references: Chen et al., BMC Bioinformatics 2013; Xie et al., Current Protocols 2021; Szklarczyk et al., Nucleic Acids Research 2025.